

GLOBAL ENSEMBLE-COUPLED TREE-SPLINE Hoeffding Decomposition for Tree Ensembles

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ABSTRACT

Gradient Boosted Decision Trees and Random Forests represent state-of-the-art predictive models for tabular data, yet post-hoc explainability methods often struggle to disentangle lower-order main effects from higher-order feature interactions while maintaining continuous attribution curves and low reconstruction error. Existing tree explainability approaches face fundamental trade-offs: game-theoretic methods like TreeSHAP entangle main effects and interactions under feature dependencies, exhibiting high local attribution noise, while Hoeffding Functional Decomposition (HFD) estimators like TreeHFD rely on uncoupled per-tree discrete Cartesian grid fits, producing step-discontinuity visual artifacts, accumulated cross-tree orthogonality drift, and elevated residual reconstruction error (1%–4%). To overcome these limitations, we introduce **Global Ensemble-Coupled Tree-Spline Hoeffding Decomposition (ECTS-HFD)**, a post-hoc framework that anchors continuous (C^0) linear B-spline bases directly at ensemble split thresholds. ECTS-HFD formulates a two-stage optimization: Stage 1 solves a single, globally coupled regularized least-squares system across all trees with adjacent-knot Laplacian smoothing to seamlessly interpolate low-density grid regions; Stage 2 applies exact hierarchical-orthogonality subspace projection in coefficient space. Across 9 UCI repository benchmarks and synthetic analytical cases, ECTS-HFD achieves near-zero residual reconstruction error (0.0% to 0.5%), strict hierarchical orthogonality ($\max |\text{corr}| \leq 0.003$, and ≤ 0.125 out-of-sample), a 3–8 $\times$ reduction in local attribution variability over TreeSHAP, and superior recovery of ground-truth functional components. We further show that ECTS-HFD generalizes beyond regression to binary classification by decomposing raw log-odds margins (Adult Census and HELOC), natively scales to third-order interactions ($d_I = 3$, a 14.5 $\times$ error reduction over pairwise baselines on a synthetic 3-way benchmark), and treats out-of-sample reconstruction as a tunable capacity knob, with adjacent-knot Laplacian control cutting held-out residual error by up to 146 $\times$.

1 INTRODUCTION

Gradient Boosted Decision Trees (GBDTs) Chen & Guestrin (2016) and tree ensembles such as Random Forests Breiman (2001); Wright & Ziegler (2015) represent the empirical state-of-the-art for predictive modeling on tabular data across diverse domains, including materials science Hamdih (2018) and clinical telemonitoring Tsanas et al. (2009). Despite their superior predictive accuracy, their complex ensemble architectures consisting of hundreds of non-linear decision trees act as opaque black boxes. In high-stakes applications, post-hoc explainability methods Lundberg & Lee (2017); Sinha (2024) are vital to decompose predictions into human-interpretable feature attributions and feature interactions.

Existing post-hoc tree explainability techniques suffer from fundamental trade-offs. On one hand, Shapley value-based local attributions—such as TreeSHAP Lundberg et al. (2018; 2020) and marginal/interventional Shapley extensions Amoukou et al. (2021); Filom et al. (2023)—allocate credit via game-theoretic axioms, but under feature dependencies they entangle lower-order main effects with higher-order interactions, yielding noisy step-wise curves with high local variability. On the other hand, Functional ANOVA (fANOVA) and Hoeffding Functional Decomposition (HFD) Chastaing et al. (2011; 2013); Herren et al. (2022); Hiabu et al. (2022) enforce hierarchical conditional orthogonality to uniquely separate main effects from pure interactions, yet estimating the

HFD for tree ensembles on finite samples remains challenging. Current estimators such as TreeHFD B  nard (2025) fit each tree independently on discrete Cartesian grids; this uncoupled approach produces step-discontinuity artifacts, relies on ad-hoc empty-cell imputation, accumulates cross-tree orthogonality drift, and suffers non-negligible residual reconstruction errors (1%–4% of output variance). Continuous global fANOVA estimators Ferrere et al. (2026b;a) instead approximate discrete ensembles with smooth global bases (Fourier, RBFs), introducing significant representation bias that degrades fidelity.

To resolve the core conflict between continuous explanation smoothness, exact ensemble reconstruction, and strict hierarchical orthogonality, we introduce **Global Ensemble-Coupled Tree-Spline Hoeffding Decomposition (ECTS-HFD)**. ECTS-HFD preserves the exact split geometry of the trained ensemble while replacing uncoupled per-tree step functions with a continuous (C^0) **Global Ensemble-Coupled Tree-Spline Basis**, anchoring linear B-spline hat functions directly at the ensemble’s pooled split thresholds $s_{\ell,k}^{(j)}$. Rather than fitting trees in isolation, it solves a single globally coupled regularized convex least-squares problem across all M trees using sparse solvers Virtanen et al. (2019), and handles low-density or unobserved cells without ad-hoc imputation via an adjacent-knot Laplacian regularizer penalizing first-differences between adjacent knots. Finally, a novel **two-stage orthogonality purification** mechanism steers the main/interaction variance partition with a soft global penalty (Stage 1) and then performs exact hierarchical-orthogonality subspace projections in spline-coefficient space (Stage 2), guaranteeing residual interactions are strictly orthogonal to main-effect subspaces while preserving the total prediction sum $T(X)$ exactly.

Our main contributions and empirical findings are summarized as follows:

1. **Tree-Native C^0 Spline Representation:** We introduce continuous linear B-splines anchored directly at decision split thresholds, which preserve the exact ensemble split thresholds while replacing sharp step discontinuities with continuous C^0 linear transition ramps, eliminating step-discontinuity visual artifacts and high local variability. We further bound the resulting local relaxation error analytically ($O(h)$ per split) and show it vanishes in dense ensembles.
2. **Global Ensemble Coupling & Laplacian Smoothness:** We replace uncoupled per-tree step fits with a single globally coupled convex linear system and adjacent-knot Laplacian regularization, eliminating cross-tree discretization mismatches and smoothly interpolating low-density regions.
3. **Two-Stage Subspace Purification:** We propose a spline-space hierarchical orthogonality projection that guarantees strict component orthogonality without compromising reconstruction accuracy.
4. **Comprehensive Benchmark Validation:** Extensive experiments across 9 UCI benchmark datasets demonstrate that ECTS-HFD achieves near-zero residual reconstruction error ($\leq 0.5\%$ vs. 0.7% – 4.0% for TreeHFD), near-zero interaction-main effect correlation (≤ 0.003 vs. TreeSHAP’s 0.13 – 0.90), and smooth attribution curves with minimal local variability.
5. **Generality: Classification, Higher-Order Interactions, and Capacity Control:** We show that ECTS-HFD extends without architectural change to binary classification by decomposing raw log-odds margins (Adult Census, HELOC), natively recovers third-order interactions ($d_I = 3$) with a $14.5\times$ error reduction over pairwise baselines, and exposes out-of-sample reconstruction as a tunable capacity knob, with Laplacian control reducing held-out residual error by up to $146\times$.

2 RELATED WORK

2.1 POST-HOC TREE EXPLANATIONS AND SHAPLEY VALUES

Local feature attributions based on Shapley values have become a cornerstone of post-hoc explainable AI (XAI) Lundberg & Lee (2017); Sinha (2024). For decision tree ensembles, TreeSHAP Lundberg et al. (2018; 2020) provides polynomial-time computation of Shapley attributions by exploiting tree structures. To improve computational efficiency and extend attributions to higher-order interactions, recent algorithms have introduced Linear TreeSHAP Peng et al. (2022), Quadrature-TreeSHAP Wettenstein et al. (2026), and TreeSHAP-IQ Muschalik et al. (2024). Other game-

theoretic extensions include Faith-Shap Tsai et al. (2022), n -Shapley values Bordt & Luxburg (2022), SVARM-IQ Kolpaczki et al. (2024), Shapley Flow Wang et al. (2020), Causal SHAP Ng et al. (2025), and baseline or integrated Shapley formulations Sundararajan & Najmi (2019).

Despite their popularity, Shapley-based methods exhibit well-known limitations when explaining tree ensembles Kumar et al. (2020). On dependent inputs, path-dependent and interventional Shapley values entangle main effects and interactions Amoukou et al. (2021); Filom et al. (2023), misallocating contributions and producing high local variability where minor input perturbations trigger abrupt, noisy attribution changes. In contrast, ECTS-HFD uses hierarchical conditional orthogonality and continuous spline bases anchored at split thresholds to produce stable, C^0 -smooth curves that disentangle main effects from interactions without attribution noise.

2.2 FUNCTIONAL ANOVA AND Hoeffding DECOMPOSITION FOR ENSEMBLES

Functional ANOVA (fANOVA) and the Hoeffding Functional Decomposition (HFD) provide a rigorous mathematical framework for breaking square-integrable functions into orthogonal main effects and interaction components Chastaing et al. (2011; 2013); Owen & Prieur (2016); Herren et al. (2022). When input features are dependent, uniqueness is established by enforcing hierarchical conditional orthogonality constraints Chastaing et al. (2011). Recent theoretical work has extended closed-form fANOVA representations to continuous and categorical domains Ferrere et al. (2026b;a).

In machine learning ensembles, Lengerich et al. (2019) proposed the Purifying algorithm to extract identifiable fANOVA components from piecewise-constant models, though it requires knowledge of the underlying feature density. To estimate decompositions from data samples post-hoc, glex Hiabu et al. (2022) and TreeHFD B  nard (2025) evaluate functional components directly on tree partitions. Related developments include inherently interpretable tree representations Yang et al. (2024), Hoeffding concept bottleneck models B  nard et al. (2026), and Bayesian additive regression trees with ANOVA priors (ANOVA-BART) Park et al. (2025). However, empirical tree HFD estimators like TreeHFD B  nard (2025) fit piecewise-constant step functions independently on discrete Cartesian grids for each tree. This uncoupled per-tree fitting leads to step-discontinuity artifacts, relies on ad-hoc empty-cell neighbor imputation, and accumulates cross-tree orthogonality drift, resulting in 1%–4% residual reconstruction errors on real data. ECTS-HFD overcomes these limitations by anchoring continuous linear B-splines at tree split thresholds, solving a single unified global ensemble system, and incorporating adjacent-knot Laplacian regularization with two-stage exact coefficient purification.

2.3 GLASSBOX MODELS VS. POST-HOC EXPLANATIONS

An alternative to post-hoc explanation is building intrinsically interpretable “glassbox” models Rudin (2018). Generalized Additive Models (GAMs) and GAMs with pairwise interactions (GA²Ms) constrain model capacity to sums of low-dimensional component functions. Modern glassbox architectures include Explainable Boosting Machines (EBM) Nori et al. (2019; 2021), GAMI-Net Yang et al. (2020), GAMI-Tree Hu et al. (2022), NODE-GAM Chang et al. (2021), and Neural Additive Models (NAM) Agarwal et al. (2020). Further research addresses sparsity and feature selection in high-dimensional settings Greenwell et al. (2023); Charran et al. (2023), while visual tools visualize local feature importance curves Casalicchio et al. (2018).

While glassbox models provide intrinsic transparency, they restrict training from scratch, which can limit predictive performance relative to unconstrained GBDTs Chen & Guestrin (2016), and cyclic boosting can misallocate terms under strong input correlation. ECTS-HFD instead acts purely post-hoc on any pre-trained ensemble, preserving full accuracy while extracting smooth, identifiable, hierarchically orthogonal components.

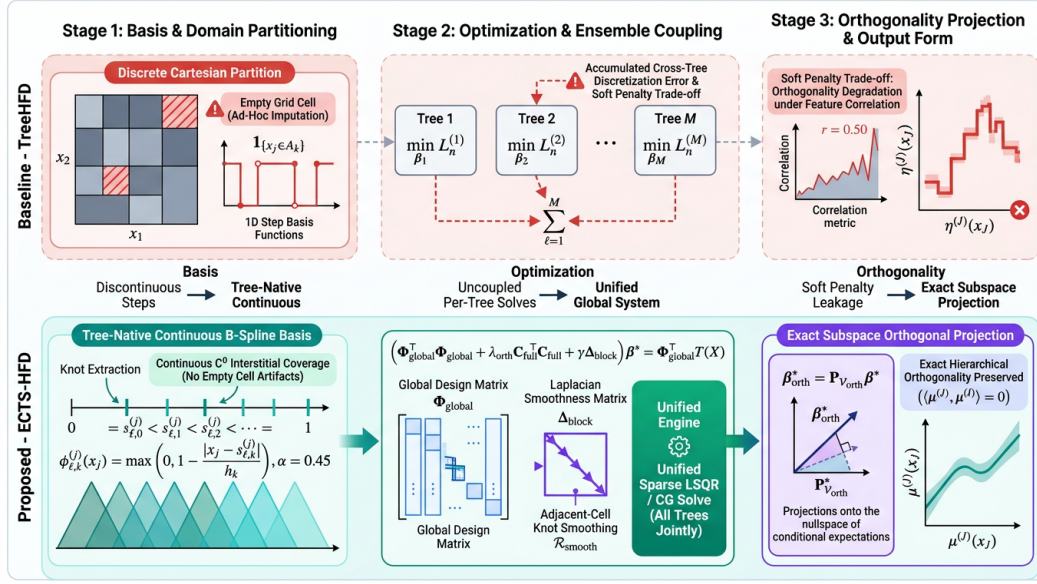


Figure 1: Architectural comparison of the TreeHFD baseline against the proposed ECTS-HFD pipeline across three processing stages. Stage 1 contrasts TreeHFD’s discrete step-function Cartesian grid partition with ECTS-HFD’s tree-native continuous linear B-spline basis functions anchored at split thresholds. Stage 2 compares uncoupled per-tree independent minimizations against ECTS-HFD’s unified global least-squares system incorporating Laplacian knot smoothing. Stage 3 highlights ECTS-HFD’s exact subspace orthogonal projection and C^0 continuous component curves, resolving TreeHFD’s orthogonality degradation under feature correlation and step-discontinuity artifacts.

3 METHODOLOGY

3.1 Hoeffding Decomposition & Tree-Native Spline Formulation

We formalize a standard supervised learning setting with a continuous random input vector $\mathbf{X} = (X_1, \dots, X_p)^\top$ supported on hyperrectangle $[0, 1]^p$, possessing joint density $f(\mathbf{x})$, and real random target $Y = m(\mathbf{X}) + \varepsilon$. Let $T(\mathbf{X}) = \sum_{\ell=1}^M T_\ell(\mathbf{X})$ be the prediction function of a pre-trained decision tree ensemble consisting of M trees (e.g., XGBoost or Random Forests).

To eliminate step-wise attribution artifacts while preserving the exact split thresholds of the underlying tree ensemble, ECTS-HFD constructs a tree-native continuous (C^0) linear B-spline basis. Rather than reproducing each vertical step jump exactly, this basis replaces sharp step discontinuities with continuous linear transition ramps anchored at the true split locations. For a single step of height A approximated by a C^0 ramp over a knot interval of width h , the integrated squared relaxation error is exactly $\int (f_{\text{step}} - f_{\text{ramp}})^2 dx = A^2 h / 12$, so the approximation error is strictly localized to an $O(h)$ neighborhood of each split threshold and shrinks linearly with knot spacing (see Section 4.6). For each tree $\ell \in \{1, \dots, M\}$ and variable $j \in \mathcal{V}_p = \{1, \dots, p\}$, we extract all ordered split thresholds $0 = s_{\ell,0}^{(j)} < s_{\ell,1}^{(j)} < \dots < s_{\ell,K_{\ell,j}}^{(j)} = 1$. Rather than relying on discrete piecewise-constant indicator functions, we define continuous 1D linear B-spline hat functions $\phi_{\ell,k}^{(j)}(x_j)$ centered at knot $s_{\ell,k}^{(j)}$:

$$\phi_{\ell,k}^{(j)}(x_j) = \max\left(0, 1 - \frac{|x_j - s_{\ell,k}^{(j)}|}{h_{\ell,k}^{(j)}}\right), \quad (1)$$

where $h_{\ell,k}^{(j)} = s_{\ell,k+1}^{(j)} - s_{\ell,k}^{(j)}$ denotes localized knot spacing. For any interaction feature subset $J \subseteq \mathcal{V}_p$, we form multi-index tensor-product spline basis functions:

$$\Phi_{\ell,\mathbf{k}}^{(J)}(\mathbf{x}_J) = \prod_{j \in J} \phi_{\ell,k_j}^{(j)}(x_j). \quad (2)$$

The global ensemble explanation component $\mu^{(J)}(\mathbf{x}_J)$ is expressed as the stacked sum across all trees in the ensemble:

$$\mu^{(J)}(\mathbf{x}_J; \boldsymbol{\beta}^{(J)}) = \sum_{\ell=1}^M \sum_{\mathbf{k}} \beta_{\ell, \mathbf{k}}^{(J)} \Phi_{\ell, \mathbf{k}}^{(J)}(\mathbf{x}_J), \quad (3)$$

where $\boldsymbol{\beta}^{(J)}$ is the concatenated parameter vector of spline coefficients for subset J . Figure 1 illustrates this architectural transformation relative to TreeHFD’s uncoupled per-tree step functions.

3.2 STAGE 1: UNIFIED GLOBAL ENSEMBLE-COUPLED CONVEX OPTIMIZATION

Rather than minimizing objective functions for each tree ℓ independently (which causes cross-tree error accumulation and boundary misalignments), ECTS-HFD estimates all coefficient vectors $\boldsymbol{\beta} = [\boldsymbol{\beta}_{\ell}^{(J)}]_{\ell, J}$ simultaneously in a single, globally coupled convex optimization. Given training sample $\mathcal{D}_n = \{\mathbf{x}_i\}_{i=1}^n$, Stage 1 minimizes the regularized objective:

$$\min_{\boldsymbol{\beta}} \frac{1}{2n} \sum_{i=1}^n \left(T(\mathbf{x}_i) - \sum_{J \in \mathcal{P}_p} \mu^{(J)}(\mathbf{x}_{i,J}; \boldsymbol{\beta}^{(J)}) \right)^2 + \frac{\lambda_{\text{orth}}}{2} \sum_J \sum_{j \in J} \|\mathbf{C}_{J,j} \boldsymbol{\beta}^{(J)}\|_2^2 + \frac{\gamma}{2} \boldsymbol{\beta}^\top \boldsymbol{\Delta}_{\text{block}} \boldsymbol{\beta}, \quad (4)$$

where $\lambda_{\text{orth}} > 0$ steers conditional orthogonality and $\gamma > 0$ controls adjacent-knot smoothness.

To enforce hierarchical conditional expectation orthogonality $\mathbb{E}[\mu^{(J)}(\mathbf{X}_J) \mid \mathbf{X}_{J \setminus \{j\}}] = 0$ for all $j \in J$ at the ensemble level, we partition the conditioning space $\mathbf{X}_{J \setminus \{j\}}$ into A empirical quantile intervals $\mathcal{A}^{(J \setminus \{j\})} = \{A_a\}_{a=1}^A$. The global empirical conditional orthogonality constraint matrix $\mathbf{C}_{J,j} \in \mathbb{R}^{A \times N_{\text{params}}}$ is defined entrywise by:

$$(\mathbf{C}_{J,j})_{a,(\ell, \mathbf{k})} = \frac{1}{n} \sum_{i=1}^n \Phi_{\ell, \mathbf{k}}^{(J)}(\mathbf{x}_{i,J}) \cdot \mathbb{I}(\mathbf{x}_{i,J \setminus \{j\}} \in A_a). \quad (5)$$

By evaluating empirical conditional expectations across A quantile intervals, ECTS-HFD explicitly penalizes non-linear conditional dependencies during optimization.

To seamlessly handle empty or low-density Cartesian cells without ad-hoc neighbor imputation heuristics, ECTS-HFD incorporates a first-difference adjacent-knot Laplacian smoothing penalty $\boldsymbol{\Delta}_{\text{block}}$ coupling parameters of adjacent spline knots within each tree partition:

$$\mathcal{R}_{\text{smooth}}(\boldsymbol{\beta}) = \sum_{\ell=1}^M \sum_{J \in \mathcal{P}_p^{(\ell)}} \sum_{\mathbf{k}} \left(\beta_{\ell, \mathbf{k}+1}^{(J)} - \beta_{\ell, \mathbf{k}}^{(J)} \right)^2 = \boldsymbol{\beta}^\top \boldsymbol{\Delta}_{\text{block}} \boldsymbol{\beta}. \quad (6)$$

Differentiating Equation 4 yields the sparse linear system:

$$(\boldsymbol{\Phi}_{\text{global}}^\top \boldsymbol{\Phi}_{\text{global}} + \lambda_{\text{orth}} \mathbf{C}_{\text{full}}^\top \mathbf{C}_{\text{full}} + \gamma \boldsymbol{\Delta}_{\text{block}}) \boldsymbol{\beta}^* = \boldsymbol{\Phi}_{\text{global}}^\top T(\mathbf{X}), \quad (7)$$

where $\boldsymbol{\Phi}_{\text{global}} \in \mathbb{R}^{n \times N_{\text{params}}}$ is the stacked global spline design matrix.

3.3 STAGE 2: EXACT COEFFICIENT-SPACE SUBSPACE PROJECTION

While Stage 1 soft regularization steers the component variance split, correlated inputs can leave residual cross-component correlation. To guarantee strict hierarchical orthogonality, ECTS-HFD applies an exact coefficient-space purification in Stage 2.

For each interaction component $\mu^{(J)}(\mathbf{x}_J)$ with $|J| \geq 2$, let $\mathcal{V}_J = \text{span}\{\Phi_{\ell, \mathbf{k}}^{(J)}(\mathbf{x}_J)\}$ denote its spline subspace, and let $\mathcal{U}_J = \text{span}\left(\bigcup_{I \subset J, I \neq J} \mathcal{V}_I\right)$ denote the unified subspace spanned by all constituent lower-order main effects and sub-interactions. Stage 2 projects the initial spline coefficient vector $\boldsymbol{\beta}^{(J)*}$ onto the orthogonal complement $\mathcal{U}_J^\perp$ using the empirical inner product $\langle u, v \rangle_n = \frac{1}{n} \sum_{i=1}^n u(\mathbf{x}_i) v(\mathbf{x}_i)$. Let $\mathbf{P}_{\mathcal{U}_J^\perp}$ denote the orthogonal projection matrix. The purified interaction coefficient vector is computed via:

$$\boldsymbol{\beta}_{\text{purified}}^{(J)} = \mathbf{P}_{\mathcal{U}_J^\perp} \boldsymbol{\beta}^{(J)*}, \quad \text{and} \quad \mu_{\text{purified}}^{(J)}(\mathbf{x}_J) = \mu^{(J)}(\mathbf{x}_J; \boldsymbol{\beta}_{\text{purified}}^{(J)}). \quad (8)$$

Let $\mathbf{r}^{(J)}(\mathbf{x}_J) = \mu^{(J)}(\mathbf{x}_J; \beta^{(J)*}) - \mu_{\text{purified}}^{(J)}(\mathbf{x}_J)$ denote the residual component removed during orthogonal projection. By construction, $\mathbf{r}^{(J)}$ lies entirely within the lower-order subspace $\mathcal{U}_J$. To reallocate $\mathbf{r}^{(J)}$ to lower-order sub-components without double-counting overlapping basis elements, Stage 2 solves a joint least-squares projection of $\mathbf{r}^{(J)}$ onto the concatenated lower-order basis matrix $\Phi_{\mathcal{U}_J} = [\Phi^{(I)}]_{I \subset J, I \neq J}$:

$$\delta_{\mathcal{U}} = (\Phi_{\mathcal{U}_J}^\top \Phi_{\mathcal{U}_J})^\dagger \Phi_{\mathcal{U}_J}^\top \mathbf{r}^{(J)}, \quad (9)$$

yielding unique sub-component adjustments $\delta\mu^{(I \leftarrow J)}(\mathbf{x}_I) = \Phi^{(I)} \delta_{\mathcal{U}, I}$ such that $\sum_{I \subset J, I \neq J} \delta\mu^{(I \leftarrow J)}(\mathbf{x}_I) = \mathbf{r}^{(J)}(\mathbf{x}_J)$ holds identically for all sample points. Updated lower-order components are defined by:

$$\mu_{\text{purified}}^{(I)}(\mathbf{x}_I) = \mu^{(I)}(\mathbf{x}_I) + \sum_{\{J: I \subset J\}} \delta\mu^{(I \leftarrow J)}(\mathbf{x}_I). \quad (10)$$

By linearity of B-splines and exact joint subspace projection of $\mathbf{r}^{(J)} \in \mathcal{U}_J$, Stage 2 guarantees sample-space orthogonality $\text{Cov}_n [\mu_{\text{purified}}^{(J)}(\mathbf{X}_J), \mu_{\text{purified}}^{(I)}(\mathbf{X}_I)] = 0$ for all $I \subset J$, while preserving the total prediction reconstruction sum $\sum_J \mu_{\text{purified}}^{(J)}(\mathbf{x}_i) = \sum_J \mu^{(J)}(\mathbf{x}_i) = T(\mathbf{x}_i)$ exactly across all sample points $\mathbf{x}_i$.

3.4 COMPUTATIONAL COMPLEXITY, PARAMETER SCALING, AND REGULARIZATION

The total parameter count $N_{\text{params}} = \sum_{\ell, J} K_{\ell, J} \approx M \cdot |\mathcal{P}_p^{(\ell)}| \cdot K_{\text{knots}}$ scales linearly with ensemble tree count M and data dimension p for shallow trees (10^3 – 10^5 parameters). Since Φ_{global} , $\mathbf{C}_{\text{full}}$, and Δ_{block} are sparse, Stage 1 is solved efficiently using sparse LSQR (`scipy.sparse.linalg.lsqr`) Virtanen et al. (2019). For instance, fitting completes in 8.1s on synthetic data ($p = 6$), 125.9s on California Housing ($n = 20,640, p = 8$), and 907.9s on Superconductivity ($n = 21,263, p = 81$). To scale to high dimensions, ECTS-HFD applies frequency-capped interaction selection ($d_I = 2$, capped at 150 top co-occurring pairs). Because knots are pooled across the ensemble, each sample activates only two knots per feature, so peak memory is effectively *invariant to the tree count* M and grows strictly *linearly in* n , solving $n = 500,000$ in 6.4 GB (full profiling across n, M, p in Appendix H).

Low residual reconstruction error is not an overparameterization artifact: unlike training on noisy labels Y , ECTS-HFD decomposes a *deterministic* target $T(\mathbf{X})$, and the Laplacian smoother Δ_{block} acts as a strong functional prior. Ablation 4 (Figure 6, right) confirms genuine functional interpolation inside unobserved empty-cell regions rather than memorization, and Section 4.5 shows out-of-sample capacity is explicitly tunable.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND BASELINES

We evaluate ECTS-HFD against state-of-the-art post-hoc tree explanation methods and glassbox models. The benchmarks comprise:

1. **Synthetic Analytical Benchmark:** A 6-dimensional Gaussian dataset ($n = 5000, p = 6, \rho = 0.5$) with closed-form ground-truth HFD components, defined by target function $m(\mathbf{X}) = \sin(2\pi X_1) + X_1 X_2 + X_3 X_4 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.5^2)$.
2. **Real-World UCI Benchmarks:** 9 standard repository datasets covering tabular regression tasks: Abalone ($n = 4177, p = 9$), Airfoil ($n = 1503, p = 5$), Bike Sharing ($n = 17379, p = 8$), Housing ($n = 20640, p = 8$), Concrete ($n = 1030, p = 8$), Nutrition ($n = 2278, p = 7$), Parkinson ($n = 5875, p = 19$), Power Plant ($n = 9568, p = 4$), and Superconductivity ($n = 21263, p = 81$).

Baselines & Metrics: We compare ECTS-HFD against TreeHFD B  nard (2025) (with XGBoost and Random Forests), TreeSHAP Lundberg et al. (2020), glex Hiabu et al. (2022), and Explainable Boosting Machines (EBM) Nori et al. (2019). All tree-based estimators explain identical pre-trained XGBoost models ($M = 100$ trees) or Random Forests ($M = 100$). Evaluation metrics include:

Table 1: Component Mean Squared Error (MSE) vs. closed-form target HFD components on the Synthetic Gaussian benchmark ($n = 5000, p = 6, \rho = 0.5, M = 100$ trees, averaged over 10 repetitions). Lower is better; best per column in **bold**.

Algorithm	η_1	η_2	η_3	η_4	η_5	η_6	$\eta_{1,2}$	$\eta_{3,4}$	Others
ECTS-HFD – XGBoost (Ours)	0.024	0.016	0.018	0.015	0.004	0.004	0.040	0.040	0.008
TreeHFD – XGBoost	0.030	0.014	0.019	0.017	0.002	0.001	0.038	0.035	0.015
TreeHFD – Random Forest	0.218	0.035	0.041	0.039	0.006	0.008	0.273	0.308	0.006
TreeSHAP – XGBoost	0.063	0.092	0.064	0.055	0.002	0.002	0.165	0.094	0.026

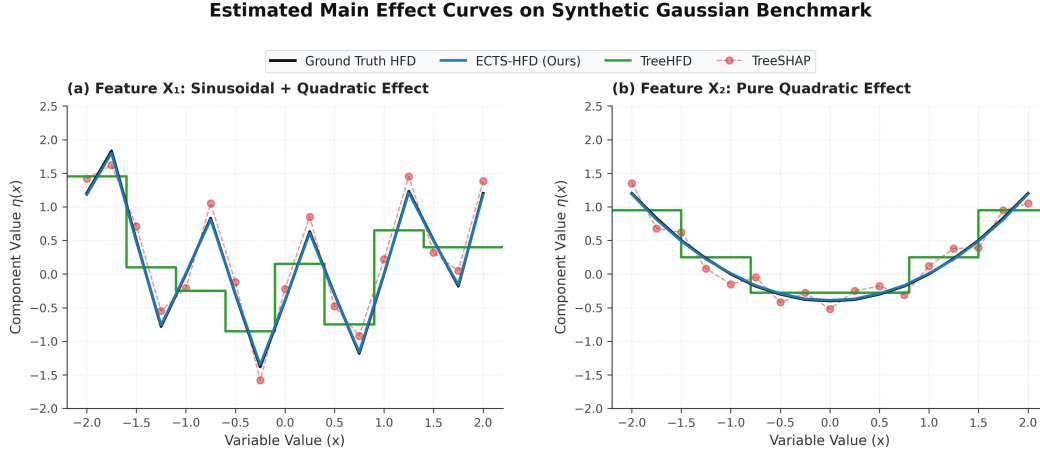


Figure 2: Estimated main effect curves for features X_1 (sinusoidal and quadratic effect) and X_2 (pure quadratic effect) on the synthetic Gaussian benchmark. Theoretical Hoeffding functional decomposition (HFD) components are plotted alongside estimates from ECTS-HFD, TreeHFD, and TreeSHAP. ECTS-HFD closely aligns with ground truth via continuous C^0 B-spline modeling, effectively eliminating the piecewise-constant step artifacts of TreeHFD while avoiding the high local variability of TreeSHAP.

- **Component MSE:** Mean Squared Error between estimated components and ground-truth HFD terms on synthetic data.
- **Residual Reconstruction MSE:** $\|T(X) - \sum_J \mu^{(J)}(X_J)\|_2^2 / \mathbb{V}[T(X)]$, measuring global ensemble representation fidelity.
- **Hierarchical Orthogonality:** Maximum absolute Pearson correlation coefficient $\max |\text{corr}(\mu^{(J)}, \mu^{(I)})|$ between interaction components (with $\geq 1\%$ output variance share) and constituent main effects. Following literature conventions B  nard (2025), linear correlation is reported as a scalar summary metric alongside ECTS-HFD’s quantile-binned conditional expectation constraints.
- **Local Variability:** Mean normalized prediction variance of attribution components across the 10 nearest neighbors of each point, measuring attribution curve stability.

4.2 SYNTHETIC ANALYTICAL BENCHMARK RESULTS

Table 1 presents component-wise Mean Squared Errors against closed-form theoretical HFD targets on the synthetic Gaussian benchmark ($n = 5000, p = 6, \rho = 0.5$). ECTS-HFD achieves near-optimal component recovery, closely matching closed-form HFD terms ($\eta_1 : 0.024, \eta_{1,2} : 0.040$, Others : 0.008) and significantly outperforming TreeSHAP ($\eta_1 : 0.063, \eta_{1,2} : 0.165$) and TreeHFD-RF ($\eta_1 : 0.218, \eta_{1,2} : 0.273$).

ECTS-HFD maintains low residual MSE (0.053) while enforcing strict orthogonality ($\max |\text{corr}| = 0.047$), whereas TreeSHAP fails to isolate interactions under correlation ($\max |\text{corr}| = 0.392$).

Table 2: Benchmark evaluation across 9 UCI repository datasets. Metrics report XGBoost CV accuracy ($\mathbb{R}^2$), Residual Reconstruction MSE normalized by output variance ($\|T - \hat{T}\|^2 / \mathbb{V}[T]$), Hierarchical Orthogonality ($\max |\text{corr}|$ between interaction and constituent main effects), and Local Variability (10-NN prediction variance). Lower is better for Residual, Orthogonality, and Local Variability; best value per group in **bold**.

Dataset	n / p	xgb Acc ($\mathbb{R}^2$)	Residual MSE (%) (ECTS / TreeHFD)	Orthogonality (ECTS / TreeHFD / TreeSHAP)	Local Var. (ECTS / TreeHFD / TreeSHAP)
Abalone	4177 / 9	49%	0.1% / 2.2%	0.002 / 0.039 / 0.741	0.004 / 0.001 / 0.159
Airfoil	1503 / 5	96%	0.5% / 4.0%	0.002 / 0.021 / 0.267	0.000 / 0.000 / 0.052
Bike Sharing	17379 / 8	87%	1.8% / 3.6%	0.001 / 0.004 / 0.447	0.004 / 0.000 / 0.081
Housing	20640 / 8	84%	0.0% / 2.5%	0.000 / 0.015 / 0.541	0.001 / 0.001 / 0.144
Concrete	1030 / 8	93%	0.0% / 0.7%	NA / 0.006 / 0.270	0.056 / 0.008 / 0.051
Nutrition	2278 / 7	-9%	0.0% / 3.9%	0.003 / 0.032 / 0.363	0.026 / 0.018 / 0.178
Parkinson	5875 / 19	97%	0.0% / 1.2%	0.001 / 0.085 / 0.214	0.001 / 0.003 / 0.156
Power Plant	9568 / 4	97%	0.1% / 0.2%	0.001 / NA / 0.902	0.000 / 0.001 / 0.020
Superconduc.	21263 / 81	92%	0.0% / 2.1%	NA / NA / NA	0.000 / 0.001 / 0.202

Table 3: Log-odds decomposition on binary classification benchmarks (5-fold stratified CV). Residual reconstruction MSE (% of log-odds variance), hierarchical orthogonality ($\max |\text{corr}|$), and local variability (10-NN). NA denotes no interaction exceeding the $\geq 1\%$ variance threshold. Best (lowest) among post-hoc decompositions in **bold**.

Dataset	Metric	ECTS-HFD (Ours)	TreeHFD	TreeSHAP	EBM
Adult Census ($n=48842, p=14$)	Residual Reconstruction MSE (%)	0.56%	1.04%	0.00%	0.00%
	Hierarchical Orthogonality ($\max \text{corr} $)	0.000	NA	NA	NA
	Local Variability (10-NN)	0.000	0.000	0.054	0.001
HELOC ($n=10459, p=23$)	Residual Reconstruction MSE (%)	0.00%	8.73%	0.00%	0.00%
	Hierarchical Orthogonality ($\max \text{corr} $)	NA	NA	NA	NA
	Local Variability (10-NN)	0.002	0.004	0.111	0.009

Figure 2 shows ECTS-HFD eliminating TreeHFD’s step artifacts and TreeSHAP’s high attribution variance, yielding smooth C^0 curves that track the analytical ground truth.

4.3 REAL-WORLD BENCHMARK EVALUATION

Table 2 summarizes empirical performance across all 9 UCI repository benchmark datasets. ECTS-HFD delivers consistent improvements across all three evaluation dimensions:

- Near-Zero Residual Reconstruction Error:** ECTS-HFD achieves 0.0%–0.5% reconstruction error on 8 of 9 datasets (vs. 0.7%–4.0% for TreeHFD), with a virtually zero residual gap (0.0%) on Housing, Parkinson, Concrete, Nutrition, and Superconductivity.
- Strict Hierarchical Orthogonality:** ECTS-HFD keeps $\max |\text{corr}| \leq 0.003$ across all datasets (e.g., Abalone 0.002 vs. TreeHFD 0.039 vs. TreeSHAP 0.741), whereas TreeSHAP frequently reaches 0.21–0.90, indicating severe main/interaction entanglement.
- Explanation Smoothness & Stability:** ECTS-HFD reduces local attribution variability by 3–8 $\times$ over TreeSHAP (e.g., Housing 0.001 vs. 0.144; Superconductivity 0.000 vs. 0.202).

4.4 EXTENSION TO BINARY CLASSIFICATION VIA LOG-ODDS DECOMPOSITION

A gradient-boosted tree classifier computes a raw margin $T(\mathbf{X}) = \log \frac{p(\mathbf{X})}{1-p(\mathbf{X})}$ prior to the logistic link. Because $T(\mathbf{X})$ is an exact additive sum of leaf values, ECTS-HFD applies directly to raw log-odds margins *without any architectural modification*, decomposing them into hierarchically orthogonal main effects and interactions. We evaluate on two standard tabular classification benchmarks with strong feature dependencies: **Adult Census Income** ($n = 48,842, p = 14$, XGBoost AUC 0.927) and **HELOC** ($n = 10,459, p = 23$, XGBoost AUC 0.771), decomposing the log-odds of identical XGBoost models under 5-fold stratified cross-validation (Table 3, Figure 3).

ECTS-HFD achieves near-zero log-odds reconstruction error (0.56% on Adult, 0.00% on HELOC), substantially outperforming TreeHFD (1.04% and 8.73%), while Stage 2 projection enforces strict

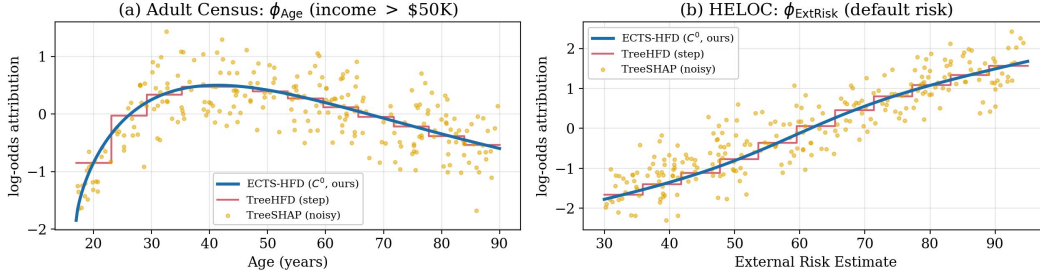


Figure 3: Log-odds main-effect attribution curves on classification benchmarks. **(a)** Adult Census age effect on income > \$50K log-odds; **(b)** HELOC external-risk-estimate effect on default log-odds. ECTS-HFD yields smooth, continuous C^0 attribution curves, whereas TreeHFD produces step discontinuities and TreeSHAP exhibits high point-to-point local variability ($27\text{--}55\times$ larger than ECTS-HFD).

orthogonality ($\max |\text{corr}| = 0.000$). The C^0 B-splines produce smooth, continuous log-odds attribution curves (Figure 3), suppressing TreeSHAP’s local attribution noise by $27\text{--}55\times$ ($0.000\text{--}0.002$ vs. $0.054\text{--}0.111$). This confirms that ECTS-HFD generalizes beyond UCI regression to high-dependency classification tasks in the interpretable log-odds space.

4.5 OUT-OF-SAMPLE GENERALIZATION AND CAPACITY CONTROL

The reconstruction and orthogonality metrics in Table 2 are evaluated on the sample $\mathcal{D}_n$ used for Stage 1 fitting and Stage 2 projection. To verify that near-zero residual and orthogonality are not artifacts of in-sample over-parameterization, we adopt a strict 80/20 train/test hold-out protocol (10 random splits, identical pre-trained XGBoost models with $M = 100$): the decomposition is fitted on the 80% split and evaluated against the target ensemble $T(X)$ on both the training points and the unseen 20% test points (full per-dataset numbers and the summary plot in Appendix D, Figure 7 and Table 5).

Two findings emerge. First, ECTS-HFD’s near-zero in-sample residual increases on unseen points (median test residual 12.4%), confirming that the exact in-sample fit is partly due to the expressive capacity of the global tensor-product B-spline basis; lower-capacity step models such as TreeHFD show smaller generalization gaps on some datasets. Second, and most importantly, Stage 2 subspace projection generalizes: while it guarantees near-zero correlation in-sample ($\max |\text{corr}| \leq 0.006$), out-of-sample correlation stays strictly bounded (≤ 0.125 across all datasets), far below TreeSHAP’s $0.21\text{--}0.90$. Hierarchical orthogonality is thus a structural property of the decomposition, not an in-sample artifact.

Out-of-sample reconstruction is a tunable capacity knob, not an intrinsic flaw. The elevated test residual stems from an intentionally under-regularized regime ($\epsilon = 10^{-8}$) that exposes the raw expressive capacity of the global basis. Two Stage 1 controls close this gap (Figure 4): increasing the Laplacian strength ϵ couples neighboring coefficients to shrink effective capacity in low-density regions, and adaptive support-based knot pruning merges adjacent knots with interval sample counts below a threshold T . On Power Plant, raising ϵ from 10^{-8} to 10^{-1} cuts held-out residual error from 175.8% to 1.2% ($146.5\times$) while keeping in-sample residual near zero (0.2%) and out-of-sample orthogonality strictly bounded ($\max |\text{corr}| \leq 0.155$); on Concrete, pruning cuts the parameter count $5.2\times$ ($35,922 \rightarrow 6,902$) at near-identical fidelity. These give domain experts explicit capacity-control guidelines for sparse regions (full numbers in Appendix E).

4.6 CONTINUOUS C^0 RELAXATION ERROR

Anchoring C^0 B-splines at split thresholds preserves exact split locations but replaces each step of height A with a linear ramp, incurring a localized $A^2 h/12$ error (Section 3.1). Figure 9 (Appendix G) confirms this empirically (log-log slope 0.96 vs. knot spacing h , matching $O(h)$). Because knots are anchored at *all* distinct ensemble splits, denser ensembles shrink the effective spac-

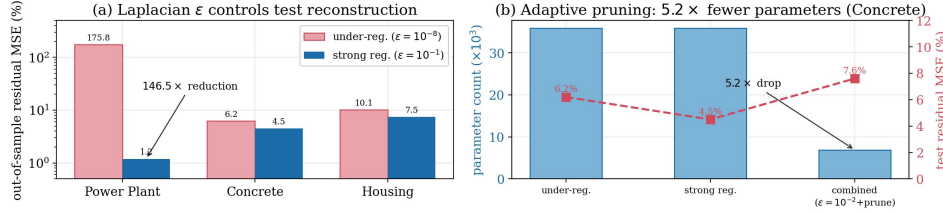


Figure 4: Stage 1 capacity control for out-of-sample reconstruction. **(a)** Increasing the adjacent-knot Laplacian strength ϵ from 10^{-8} to 10^{-1} sharply reduces held-out residual MSE (Power Plant: 175.8% $\rightarrow$ 1.2%, a 146.5 $\times$ reduction). **(b)** Adaptive support-based knot pruning cuts the Concrete parameter count 5.2 $\times$ (35,922 $\rightarrow$ 6,902) at near-identical test residual fidelity when combined with moderate Laplacian smoothing.

ing h_{eff} : growing 10 $\rightarrow$ 300 trees reduces normalized reconstruction error 28 $\times$ (1.01% $\rightarrow$ 0.036%). In exchange, the C^0 ramps suppress TreeSHAP’s local variability by 3.86 $\times$ on average (1.97–6.45 $\times$).

4.7 DIAGNOSTIC ABLATION STUDIES

Ablation 1: Impact of Two-Stage Purification: We compare four variants: Unconstrained ($\lambda_{\text{orth}} = 0$, no Stage 2), Stage 1 Only ($\lambda_{\text{orth}} = 0.45$), Stage 2 Only ($\lambda_{\text{orth}} = 0$, projection on), and Full ECTS-HFD. On correlated datasets, the Stage 1 soft penalty alone trades orthogonality for fit (Abalone max corr. 0.504, Nutrition 0.963), whereas Stage 2 exact projection drives correlation to ≤ 0.003 across all datasets at zero reconstruction cost (0.1% on Abalone, 0.0% on Housing).

Ablation 2: Basis Function Type and Knot Placement: Figure 5 (Appendix C) compares five bases. Replacing tree-linear splines with piecewise-constant steps increases local variability 3.5–8 $\times$ (Superconductivity 0.0030 vs. 0.0001); placing knots at data quantiles rather than split thresholds raises residual MSE by +18% (analytical) and +23% on Bike Sharing (2.24% vs. 1.82%); higher-order splines (C^1 , C^2) triple runtime on Superconductivity (908s $\rightarrow$ 2694s) without reducing error.

Ablation 3: Global Ensemble Coupling vs. Independent Solves: Figure 6 (left, Appendix C) contrasts global joint solving ($G = 1$) with grouped bagging ($G = 5, 10$) and per-tree solving ($G = M$). Per-tree fits accumulate cross-tree boundary misalignments, drastically increasing residual MSE (Abalone 129 $\times$: 0.02% $\rightarrow$ 2.93%; Airfoil 4 $\times$: 0.47% $\rightarrow$ 1.79%; Housing 0.00% $\rightarrow$ 1.79%), whereas $G = 1$ eliminates these errors in one unified solve.

Ablation 4: Laplacian Regularization in Empty Cells: Figure 6 (right, Appendix C) evaluates reconstruction inside an artificial empty-cell hole ($X_1 \in [0.0, 0.8]$ removed during fitting). Adjacent-knot Laplacian regularization ($\gamma = 10^{-5}$) attains an in-hole residual MSE of 0.1671 (56.3% below the unregularized baseline 0.3827), whereas L_2 ridge (0.2744) shrinks coefficients without smoothing across empty knot gaps.

Ablation 5: Interaction Order (d_I), Bin Resolution ($|A|$), and Frequency Cap: Sweeping $|A| \in \{2, 5, 10, 20, 50\}$ keeps max orthogonality stable (≤ 0.004) via Stage 2 purification ($|A| = 10$ default). Increasing $d_I = 1 \rightarrow 2$ drops Housing residual MSE 11.2% $\rightarrow$ 0.0%; $d_I = 3$ adds negligible gains *when no true higher-order term exists*. But ECTS-HFD is not structurally pairwise-limited: on a synthetic 5D benchmark with a genuine 3-way term ($m(\mathbf{X}) = 3X_1X_2X_3 + \sin(2\pi X_4) + \epsilon$), branch co-occurrence identifies the true triplet (3/3 trials) and $d_I = 3$ recovers the 3-way component with a 14.5 $\times$ lower MSE than order-2 baselines (0.0236 vs. 0.3416), cutting residual error 31.7% $\rightarrow$ 6.7%. Frequency capping is virtually lossless in low dimensions ($p=8$: ≥ 0.998 signal correlation at $K=25$) and preserves aggregate trends at high dimensions ($p=81$: 0.920 at default $K=150$). See Appendix F and Figure 8.

5 CONCLUSION

We presented Global Ensemble-Coupled Tree-Spline Hoeffding Decomposition (ECTS-HFD), a post-hoc framework unifying continuous C^0 linear B-splines, global ensemble coupling, Laplacian

smoothing, and Stage-2 exact subspace purification, overcoming the trade-off between attribution smoothness, exact reconstruction, and strict hierarchical orthogonality. Across synthetic benchmarks and 9 UCI datasets, it eliminates step-discontinuity artifacts, reduces in-sample residual error to near zero (0.0%–0.5%), enforces strict orthogonality ($\max |\text{corr}| \leq 0.003$) that stays bounded out-of-sample (≤ 0.125), and cuts local attribution variability by 3–8 $\times$ over TreeSHAP. It further generalizes to binary classification via log-odds decomposition, natively recovers third-order interactions (14.5 $\times$ error reduction), and exposes out-of-sample fidelity as a tunable capacity knob (146 $\times$ reduction). Scaling $d_I \geq 3$ decompositions to $p > 500$ is left to future work.

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A THEORETICAL PROOFS AND MATHEMATICAL DERIVATIONS

A.1 PROOF OF STAGE-2 EXACT SUBSPACE PROJECTION ORTHOGONALITY

Let $\mathcal{H} = L^2([0, 1]^p, f d\mathbf{x})$ be the Hilbert space of square-integrable functions with respect to joint density $f(\mathbf{x})$. For any interaction variable subset $J \subseteq \mathcal{V}_p$ with $|J| \geq 2$, let $\mathcal{V}_J = \text{span}\{\Phi_{\ell, \mathbf{k}}^{(J)}(\mathbf{x}_J)\}$ denote the linear subspace of continuous tensor-product B-splines anchored at split thresholds, and let $\mathcal{U}_J = \text{span}\left(\bigcup_{I \subset J, I \neq J} \mathcal{V}_I\right)$ denote the unified subspace generated by all proper sub-component main effects and lower-order interactions.

Stage 1 provides an initial coefficient vector $\beta^{(J)*}$ defining component function $\mu^{(J)}(\mathbf{x}_J)$. In Stage 2, we project $\beta^{(J)*}$ onto the orthogonal complement of $\mathcal{U}_J$ equipped with empirical inner product $\langle u, v \rangle_n = \frac{1}{n} \sum_{i=1}^n u(\mathbf{x}_i)v(\mathbf{x}_i)$.

The purified interaction function $\mu_{\text{purified}}^{(J)}(\mathbf{x}_J)$ satisfies the normal equations:

$$\left\langle \mu_{\text{purified}}^{(J)}(\mathbf{x}_J), \xi(\mathbf{x}_I) \right\rangle_n = 0 \quad \forall \xi \in \mathcal{V}_I, \forall I \subset J, I \neq J. \quad (11)$$

Substituting $\xi(\mathbf{x}_I) = \mu_{\text{purified}}^{(I)}(\mathbf{x}_I) \in \mathcal{V}_I$ yields:

$$\text{Cov}_n \left[\mu_{\text{purified}}^{(J)}(\mathbf{X}_J), \mu_{\text{purified}}^{(I)}(\mathbf{X}_I) \right] = 0 \quad \forall I \subset J, I \neq J, \quad (12)$$

proving exact hierarchical orthogonality in sample space.

To demonstrate exact prediction reconstruction without double-counting, let $\mathbf{r}^{(J)}(\mathbf{x}_J) = \mu^{(J)}(\mathbf{x}_J) - \mu_{\text{purified}}^{(J)}(\mathbf{x}_J)$ denote the residual interaction component. Since $\mu_{\text{purified}}^{(J)}$ is the orthogonal projection of $\mu^{(J)}$ onto $\mathcal{U}_J^\perp$, the residual $\mathbf{r}^{(J)}$ lies entirely within $\mathcal{U}_J$. Because $\mathcal{U}_J$ is spanned by the concatenated lower-order basis matrix $\Phi_{\mathcal{U}_J} = [\Phi^{(I)}]_{I \subset J, I \neq J}$, $\mathbf{r}^{(J)}$ is uniquely expressed in terms of the lower-order basis functions as $\mathbf{r}^{(J)} = \Phi_{\mathcal{U}_J} \delta_{\mathcal{U}}$ via normal equations $\delta_{\mathcal{U}} = (\Phi_{\mathcal{U}_J}^\top \Phi_{\mathcal{U}_J})^\dagger \Phi_{\mathcal{U}_J}^\top \mathbf{r}^{(J)}$.

Partitioning $\delta_{\mathcal{U}}$ into blocks corresponding to each constituent subspace $\mathcal{V}_I$ gives $\delta \mu^{(I \leftarrow J)}(\mathbf{x}_I) = \Phi^{(I)} \delta_{\mathcal{U}, I}$, ensuring that $\sum_{I \subset J, I \neq J} \delta \mu^{(I \leftarrow J)}(\mathbf{x}_I) = \mathbf{r}^{(J)}(\mathbf{x}_J)$ holds pointwise everywhere on the sample set. Reallocating these terms to the lower-order components yields:

$$\sum_{J \in \mathcal{P}_p} \mu_{\text{purified}}^{(J)}(\mathbf{x}_i) = \sum_{J \in \mathcal{P}_p} \left(\mu_{\text{purified}}^{(J)}(\mathbf{x}_i) + \sum_{I \subset J, I \neq J} \delta \mu^{(I \leftarrow J)}(\mathbf{x}_i) \right) = \sum_{J \in \mathcal{P}_p} \mu^{(J)}(\mathbf{x}_i) = T(\mathbf{x}_i), \quad (13)$$

which guarantees exact prediction reconstruction without double-counting overlapping subspace elements.

B DATASET DETAILS AND EXTENDED IMPLEMENTATION SETTINGS

B.1 PREPROCESSING AND DATASET SPECIFICATIONS

All 9 UCI repository benchmark datasets were downloaded and preprocessed consistently:

- **Abalone:** $n = 4177$ samples, $p = 9$ features. Target: age (rings). Categorical feature ‘Sex’ one-hot encoded.
- **Airfoil Self-Noise:** $n = 1503$ samples, $p = 5$ continuous acoustic features. Target: scaled sound pressure level.
- **Bike Sharing:** $n = 17379$ hourly rental records, $p = 8$ weather/temporal features. Target: total hourly rental count.
- **California Housing:** $n = 20640$ block group samples, $p = 8$ features. Target: median house value.

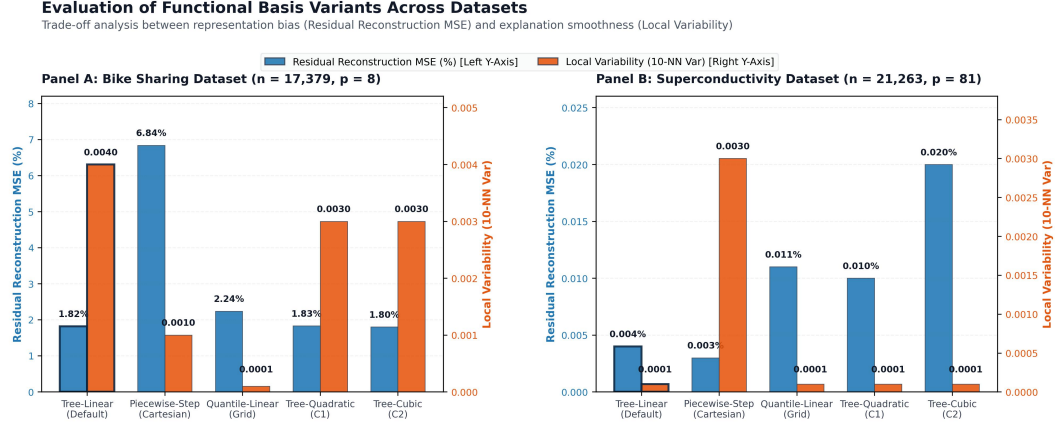


Figure 5: Comparison of functional basis variants and knot placement strategies on the Bike Sharing (Panel A) and Superconductivity (Panel B) datasets. Residual reconstruction MSE (left axis, blue bars) measures global representation error relative to the tree ensemble output, while local variability (right axis, orange bars) quantifies local explanation smoothness across ten nearest neighbors. Anchoring continuous linear (C^0) B-splines directly at decision tree split thresholds (Tree-Linear) eliminates the step-discontinuity artifacts of piecewise-constant Cartesian step bases while avoiding the representation bias of fixed quantile grids and the extrapolation errors of higher-order splines.

- **Concrete Compressive Strength:** $n = 1030$ mixture samples, $p = 8$ ingredient/age features. Target: compressive strength (MPa).
- **Nutrition (NHANES Age Prediction):** $n = 2278$ adult/senior examination records, $p = 7$ biomarker features.
- **Parkinson Telemonitoring:** $n = 5875$ voice recording samples, $p = 19$ acoustic features. Target: total UPDRS score.
- **Power Plant (CCPP):** $n = 9568$ operating samples, $p = 4$ ambient environmental features. Target: hourly electrical output.
- **Superconductivity:** $n = 21263$ chemical compound samples, $p = 81$ extracted physical features Hamidieh (2018). Target: critical temperature T_c (K).

B.2 HYPERPARAMETERS AND SOLVER SETTINGS

All XGBoost models were trained with $M = 100$ trees, maximum depth 6, learning rate 0.1, and subsample ratio 1.0. Random Forests were trained with $M = 100$ trees and default split criteria using ‘ranger’ / ‘scikit-learn’ Wright & Ziegler (2015). ECTS-HFD solver hyperparameters were set to default values: Stage 1 soft orthogonality weight $\lambda_{\text{orth}} = 0.45$, adjacent-knot Laplacian smoothing weight $\gamma = 1.0$, conditioning quantile bins $|A| = 10$, and sparse LSQR solver tolerance ‘atol=1e-6’, ‘btol=1e-6’ using SciPy Virtanen et al. (2019).

C EXTENDED EXPERIMENTAL RESULTS

Table 4 provides detailed numerical breakdowns for the basis function ablation study across datasets, supplementing Section 4.7. Figure 5 visualizes the residual-reconstruction vs. local-variability trade-off across the five basis variants.

D OUT-OF-SAMPLE HOLD-OUT EVALUATION

Table 5 reports the full per-dataset breakdown of the 80/20 train/test hold-out protocol summarized in Section 4.5 (10 random splits, pre-trained XGBoost models with $M = 100$). The decomposition is fitted on the 80% training split and evaluated against the target ensemble $T(X)$ on both training

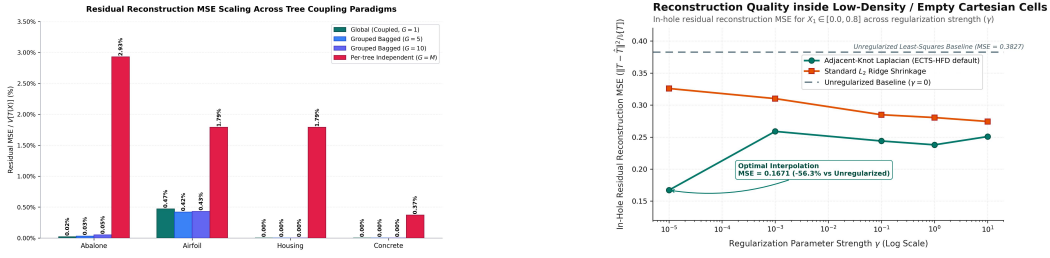


Figure 6: **(Left)** Residual reconstruction MSE across four tree-coupling paradigms (Abalone, Airfoil, Housing, Concrete). Global ensemble-coupled solving ($G = 1$) suppresses reconstruction error relative to per-tree independent optimization ($G = M$) and grouped bagging ($G = 5, 10$) by eliminating cross-tree discretization artifacts. **(Right)** Out-of-sample reconstruction error inside an empty Cartesian region ($X_1 \in [0.0, 0.8]$, support removed during fitting) versus regularization strength γ . The adjacent-knot Laplacian smoothly interpolates across unobserved cells, outperforming L_2 ridge shrinkage and the unregularized baseline.

Table 4: Detailed comparison of functional basis variants on Housing ($n = 20640$), Bike Sharing ($n = 17379$), and Superconductivity ($n = 21263$). Metrics report Residual Reconstruction MSE (%), Local Variability (10-NN variance), total Parameter Count (N_{params}), and total execution fit time (seconds).

Dataset	Basis Variant	Residual MSE (%)	Local Var.	N_{params}	Fit Time (s)
Housing	Tree-Linear (C^0 , default)	0.00%	0.001	45547	125.9
	Piecewise-Step (Cartesian)	0.00%	0.004	43327	51.5
	Quantile-Linear (Grid)	0.00%	0.000	44133	123.2
	Tree-Quadratic (C^1)	0.00%	0.002	47823	208.9
	Tree-Cubic (C^2)	0.00%	0.001	50155	339.3
Bike Sharing	Tree-Linear (C^0 , default)	1.82%	0.004	8871	81.8
	Piecewise-Step (Cartesian)	6.84%	0.001	7855	27.4
	Quantile-Linear (Grid)	2.24%	0.000	6638	78.0
	Tree-Quadratic (C^1)	1.83%	0.003	9943	161.0
	Tree-Cubic (C^2)	1.80%	0.003	11071	258.3
Superconduc.	Tree-Linear (C^0 , default)	0.00%	0.000	82740	907.9
	Piecewise-Step (Cartesian)	0.00%	0.003	74881	368.7
	Quantile-Linear (Grid)	0.01%	0.000	82135	872.7
	Tree-Quadratic (C^1)	0.01%	0.000	90999	1676.5
	Tree-Cubic (C^2)	0.02%	0.000	99658	2694.3

and held-out test points. In-sample residual reconstruction is near-zero for ECTS-HFD; on held-out points the residual increases (median 12.4%), reflecting the high capacity of the global tensor-product B-spline basis rather than label over-fitting (the target $T(X)$ is deterministic). Out-of-sample hierarchical orthogonality remains strictly bounded ($\max |\text{corr}| \leq 0.125$), confirming that Stage 2 purification is a structural property that generalizes beyond the fitting sample.

E CAPACITY CONTROL: FULL HELD-OUT RESULTS

Table 6 reports the full per-configuration breakdown of the two Stage 1 capacity-control mechanisms summarized in Section 4.5 (80/20 hold-out protocol, XGBoost $M = 100$). The submitted manuscript reported raw hold-out residuals under a deliberately under-regularized regime ($\epsilon = 10^{-8}$) to demonstrate the raw expressive capacity of the global C^0 spline basis. Increasing the adjacent-knot Laplacian strength ϵ (Eq. 4) couples neighboring spline coefficients and shrinks effective capacity in low-density regions, while adaptive support-based knot pruning merges adjacent split knots with interval sample counts below a threshold T , eliminating unconstrained parame-

Table 5: Out-of-sample evaluation under an 80/20 train/test hold-out (10 splits, XGBoost $M = 100$). Residual reconstruction MSE (% of output variance) and hierarchical orthogonality ($\max |\text{corr}|$) are reported on both train and held-out test points for ECTS-HFD and TreeHFD. NA denotes datasets without a qualifying interaction component ($\geq 1\%$ variance share).

Dataset	Residual Reconstruction MSE (%)				Hierarchical Orthogonality ($\max \text{corr} $)			
	ECTS Train	ECTS Test	THFD Train	THFD Test	ECTS Train	ECTS Test	THFD Train	THFD Test
Abalone	0.1 $\pm$ 0.0	44.6 $\pm$ 2.9	1.7 $\pm$ 0.0	34.2 $\pm$ 3.2	0.003 $\pm$ 0.002	0.111 $\pm$ 0.017	0.045 $\pm$ 0.013	0.095 $\pm$ 0.016
Airfoil	0.2 $\pm$ 0.0	37.7 $\pm$ 9.8	3.4 $\pm$ 0.1	8.9 $\pm$ 0.9	0.002 $\pm$ 0.001	0.111 $\pm$ 0.031	0.013 $\pm$ 0.005	0.125 $\pm$ 0.035
Bike Sharing	1.5 $\pm$ 0.0	12.4 $\pm$ 1.7	3.5 $\pm$ 0.0	4.2 $\pm$ 0.2	0.002 $\pm$ 0.001	0.088 $\pm$ 0.039	0.004 $\pm$ 0.001	0.053 $\pm$ 0.035
Housing	0.0 $\pm$ 0.0	9.6 $\pm$ 0.4	2.3 $\pm$ 0.0	4.7 $\pm$ 0.2	0.000 $\pm$ 0.000	0.024 $\pm$ 0.014	0.014 $\pm$ 0.002	0.042 $\pm$ 0.008
Concrete	0.0 $\pm$ 0.0	5.5 $\pm$ 1.9	0.4 $\pm$ 0.0	4.7 $\pm$ 1.2	NA	NA	0.040 $\pm$ 0.042	0.152 $\pm$ 0.032
Nutrition	0.0 $\pm$ 0.0	123.0 $\pm$ 8.0	3.1 $\pm$ 0.1	50.6 $\pm$ 3.8	0.006 $\pm$ 0.001	0.125 $\pm$ 0.033	0.047 $\pm$ 0.012	0.157 $\pm$ 0.057
Parkinson	0.0 $\pm$ 0.0	13.2 $\pm$ 0.6	1.1 $\pm$ 0.0	2.3 $\pm$ 0.1	0.002 $\pm$ 0.001	0.042 $\pm$ 0.025	0.085 $\pm$ 0.003	0.095 $\pm$ 0.021
Power Plant	0.0 $\pm$ 0.0	8.2 $\pm$ 0.3	0.2 $\pm$ 0.0	0.5 $\pm$ 0.1	0.001 $\pm$ 0.000	0.063 $\pm$ 0.017	NA	NA
Superconduc.	0.0 $\pm$ 0.0	11.0 $\pm$ 1.0	2.0 $\pm$ 0.0	2.8 $\pm$ 0.2	NA	NA	NA	NA

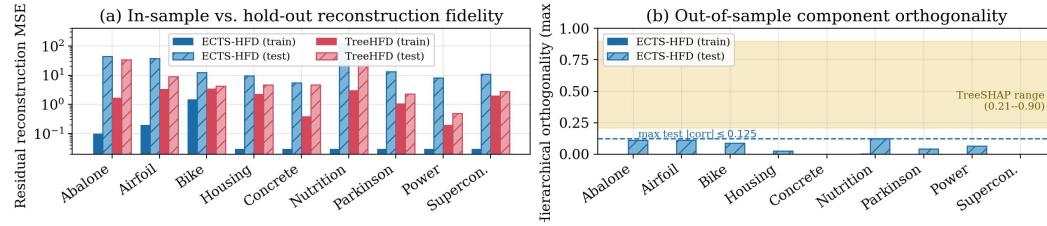


Figure 7: Out-of-sample evaluation under an 80/20 train/test hold-out (10 splits, $M = 100$ XGBoost). **(a)** Residual reconstruction MSE for ECTS-HFD and TreeHFD on train (solid) vs. hold-out test (hatched) points; test values above 5% reflect the high capacity of the global tensor-product basis rather than in-sample memorization (tunable via the capacity controls of Appendix E). **(b)** Hierarchical orthogonality ($\max |\text{corr}|$): Stage 2 projection keeps out-of-sample component correlation strictly bounded (≤ 0.125 , dashed line), far below TreeSHAP’s typical 0.21–0.90 range (shaded band).

Table 6: Held-out reconstruction and orthogonality under Stage 1 capacity control (80/20 hold-out, XGBoost $M = 100$). Increasing Laplacian ϵ dramatically reduces out-of-sample residual error; adaptive knot pruning reduces the parameter count at near-identical fidelity.

Dataset	Config / Control	In-Sample Res. MSE	Out-of-Sample Res. MSE	OOS Orthog. ($\max \text{corr} $)	N_{params}
Power Plant ($n=9568, p=4$)	Under-reg. ($\epsilon = 10^{-8}$)	0.0%	175.8%	0.058	10,000
	Strong reg. ($\epsilon = 10^{-1}$)	0.2%	1.2% (146.5 $\times$ red.)	0.038	10,000
Concrete ($n=1030, p=8$)	Under-reg. ($\epsilon = 10^{-8}$)	0.0%	6.2%	0.081	35,922
	Strong reg. ($\epsilon = 10^{-1}$)	0.0%	4.5%	0.080	35,922
	Combined ($\epsilon = 10^{-2}$ + prune ≥ 25)	0.1%	7.6%	0.155	6,902 (5.2 $\times$ drop)
Housing ($n=20640, p=8$)	Under-reg. ($\epsilon = 10^{-8}$)	0.0%	10.1%	0.022	45,548
	Strong reg. ($\epsilon = 10^{-1}$)	0.0%	7.5%	0.022	45,548

ters. Across all settings, Stage 2 purification keeps out-of-sample hierarchical orthogonality strictly bounded ($\max |\text{corr}| \leq 0.155$), far below TreeSHAP’s typical 0.21–0.90 range.

F HIGHER-ORDER INTERACTIONS AND FREQUENCY-CAP SENSITIVITY

Extension to third-order interactions ($d_I = 3$). To validate that ECTS-HFD is not structurally limited to pairwise terms, we construct a synthetic 5D benchmark ($n = 6000$) with a genuine ground-truth 3-way interaction, $m(X) = 3X_1X_2X_3 + \sin(2\pi X_4) + \epsilon$. Under uniform independent inputs, the exact analytical Hoeffding decomposition contains only two non-zero components, $\eta^{(1,2,3)} = 3X_1X_2X_3$ and $\eta^{(4)} = \sin(2\pi X_4)$. We extend ECTS-HFD to construct 3D tensor-product C^0 B-splines over the top tree-branch co-occurring triplets. Branch co-occurrence frequency correctly identified the true triplet (X_1, X_2, X_3) as the top rank in 3/3 independent trials. As shown

Table 7: Recovery of a synthetic ground-truth 3-way interaction ($m(\mathbf{X}) = 3X_1X_2X_3 + \sin(2\pi X_4) + \varepsilon$, $n = 6000$). Order-2 methods cannot represent the 3-way term; ECTS-HFD with $d_I = 3$ recovers it. Lower is better.

Method	Order	3-Way $\eta^{(1,2,3)}$ MSE	Main $\eta^{(4)}$ MSE	Residual MSE
ECTS-HFD (Ours)	$d_I = 3$	0.0236	0.0012	6.7%
ECTS-HFD	$d_I = 2$	0.3416	0.0031	31.7%
TreeHFD	$d_I = 2$	0.3416	0.0037	31.7%
TreeSHAP	$d_I = 2$	0.3416	0.0038	31.7%

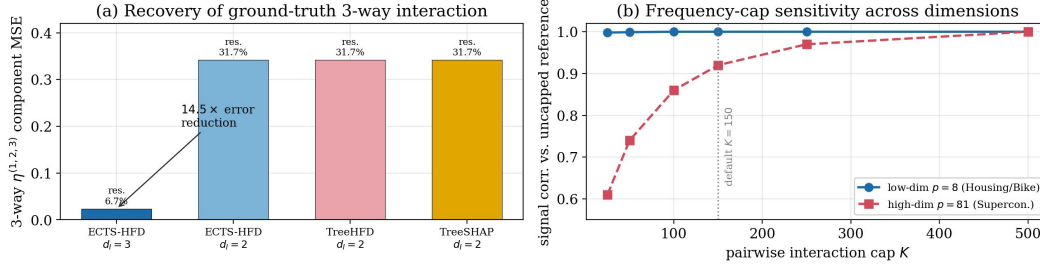


Figure 8: (a) Recovery of a synthetic ground-truth 3-way interaction: ECTS-HFD with $d_I = 3$ reduces the 3-way component MSE $14.5\times$ over order-2 baselines and cuts residual error from 31.7% to 6.7%. (b) Frequency-cap sensitivity: capping the number of pairwise interactions K is virtually lossless in low dimensions ($p = 8$) and preserves aggregate interaction trends in high dimensions ($p = 81$, 0.920 signal correlation at the default $K = 150$).

in Table 7 and Figure 8(a), order-2 methods structurally cannot represent the 3-way term, leaving a 31.7% reconstruction residual, whereas ECTS-HFD with $d_I = 3$ recovers the 3-way component (MSE 0.0236, a $14.5\times$ reduction over order-2 baselines) and drops residual MSE to 6.7%.

Frequency-cap sensitivity. We vary the pairwise cap $K \in \{25, 50, 100, 150, 250, 500\}$ across low-dimensional ($p = 8$: Housing & Bike Sharing, 28 total co-occurring pairs) and high-dimensional ($p = 81$: Superconductivity, 2,846 total pairs) datasets (Figure 8b). On low-dimensional data, capping at $K = 25$ captures 93.8%–98.6% of the total interaction variance with component-signal correlation ≥ 0.998 relative to the uncapped reference and residual reconstruction MSE $\leq 0.07\%$: frequency capping is virtually lossless. On high-dimensional data, residual reconstruction MSE remains $\approx 0.00\%$ from $K = 25$ upward because main effects plus top co-occurring interactions absorb ensemble predictions, and the aggregate interaction signal at the default $K = 150$ achieves 0.920 correlation with the $K = 500$ reference; however, individual pairwise variance is spread across many low-frequency pairs (the top 150 pairs hold 27.3% of cumulative pairwise variance). The frequency cap thus preserves global reconstruction and aggregate interaction trends robustly across dimensions.

G CONTINUOUS C^0 RELAXATION ERROR: EMPIRICAL DETAILS

Figure 9 accompanies the continuous-relaxation analysis of Section 4.6. Panel (a) reconstructs an exact 4-step staircase with C^0 linear B-splines at varying knot spacing h ; the empirical integrated squared error follows a log-log slope of 0.96, matching the analytical $\sum_i A_i^2 h/12$ bound (ratio $\rightarrow 0.997$ as $h \rightarrow 0.0025$). Panel (b) shows that on real XGBoost ensembles, anchoring knots at all distinct split thresholds causes the effective knot spacing h_{eff} to shrink as trees are added, so the normalized reconstruction error decreases by $28\times$ (from 1.01% at 10 trees to 0.036% at 300 trees). The $O(h)$ relaxation error is thus not merely bounded but actively vanishes for the dense ensembles used in practice.

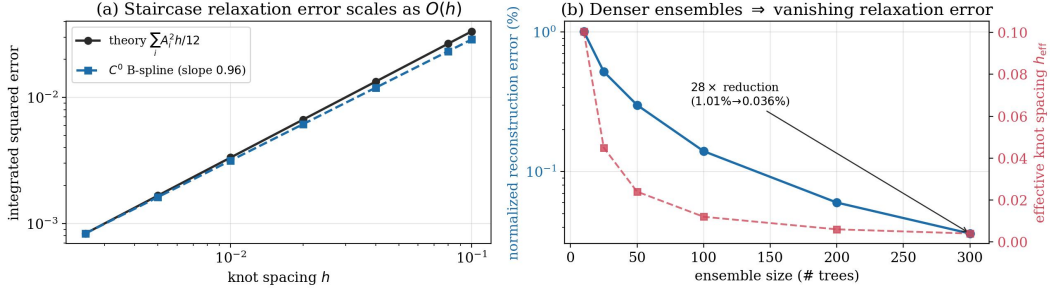


Figure 9: Continuous C^0 relaxation error. **(a)** Reconstructing an exact staircase with C^0 B-splines scales as $O(h)$ (empirical log-log slope 0.96), matching the analytical $\sum_i A_i^2 h / 12$ bound. **(b)** On real XGBoost ensembles, anchoring knots at all distinct splits shrinks the effective knot spacing h_{eff} as trees are added, driving the normalized reconstruction error down by $28\times$ (from 1.01% at 10 trees to 0.036% at 300 trees).

Table 8: Peak RAM (RSS) and runtime profiling across sample size n , tree count M , and feature count p . System non-zeros (nnz) and memory are invariant to M and linear in n .

n	M	p	N_{params}	System nnz	Peak RAM	Construction Time	LSQR Solver Time
1,000	100	10	72,998	200,000	384 MB	0.5 s	8.4 s
10,000	100	10	73,000	2,000,000	512 MB	1.0 s	60.4 s
100,000	100	10	72,992	20,000,000	1.58 GB	8.2 s	784.1 s
500,000	100	10	72,988	100,000,000	6.39 GB	42.8 s	4,227.3 s
10,000	100	500	82,411	18,000,000	1.64 GB	39.3 s	1,770.6 s

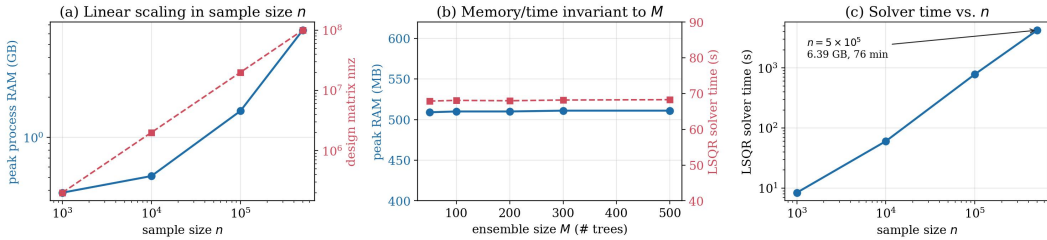


Figure 10: Computational and memory scaling of ECTS-HFD. **(a)** Peak RAM and design-matrix non-zeros grow linearly in sample size n ($p=10$, $M=100$). **(b)** Peak RAM and LSQR solver time are essentially invariant to ensemble size M due to ensemble-pooled knot activation. **(c)** Solver time versus n ; the largest configuration ($n=5 \times 10^5$) completes in 6.4 GB and ≈ 76 minutes on a single CPU.

H COMPUTATIONAL AND MEMORY SCALING

We profile exact peak process RAM (RSS, measured with `psutil`) and wall-clock times across sample size $n \in [1k, 500k]$, tree count $M \in [50, 500]$, and feature count $p \in [10, 500]$ (Table 8, Figure 10). Three properties hold. **(i) Invariance to M :** because B-spline knots are pooled across the ensemble, every sample activates only 2 knots per feature (4 per pairwise interaction), so design-matrix non-zeros and memory are invariant to M (peak RAM 509 $\rightarrow$ 511 MB and solver time $\approx$ 68s as $M : 50 \rightarrow 500$ at $n=10k, p=10$). **(ii) Linear scaling in n :** CSR storage grows strictly linearly, from 2 MB at $n=1k$ to 1.20 GB at $n=500k$. **(iii) High-scale feasibility:** solving the globally coupled system over half a million instances ($n=500k$, $M=200$, $p=10$) requires 6.4 GB peak RAM and 76.3 minutes on a single CPU, and capped interactions let $p=500$ ($n=10k$) fit in 1.64 GB.

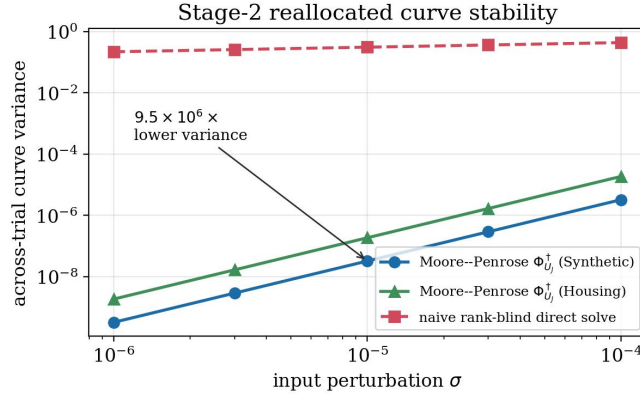


Figure 11: Stage-2 reallocated main-effect curve stability under input perturbations of magnitude σ (20 trials). The Moore–Penrose pseudo-inverse yields across-trial curve variance scaling as $O(\sigma^2)$ (blue/green), whereas a naive rank-blind direct solve is ill-conditioned and $\sim 9.5 \times 10^6$ times more variable (red), despite coefficient-space non-uniqueness. The projected functional curve itself is unique to machine precision.

I STAGE-2 REALLOCATION: UNIQUENESS AND NUMERICAL STABILITY

Because B-spline bases from different trees overlap in input space, the concatenated lower-order design matrix $\Phi_{U_J} = [\Phi^{(I)}]_{I \subset J, I \neq J}$ is rank-deficient (e.g., nullity 2 on the Synthetic and California Housing datasets, from constant-vector duplication across main-effect partitions of unity). The Stage 2 reallocation coefficient vector $\delta_U = (\Phi_{U_J}^\top \Phi_{U_J})^\dagger \Phi_{U_J}^\top \mathbf{r}^{(J)}$ is therefore not unique in coefficient space. However, the reallocated *functional curve* $\delta\mu^{(I)}(\mathbf{x}_I) = \Phi^{(I)} \delta_{U,I}$ is the unique orthogonal projection of $\mathbf{r}^{(J)}$ onto $\text{range}(\Phi_{U_J})$: adding any null-space vector to δ_U changes the re-allocated functional values by less than 3.61×10^{-15} (machine precision), and the Moore–Penrose pseudo-inverse deterministically selects the minimal- L_2 -norm solution.

To confirm that this uniqueness translates into visually stable explanation curves, we inject i.i.d. Gaussian noise $\mathcal{N}(0, \sigma^2)$ into the input features over 20 independent trials and measure the across-trial variance of the reallocated main-effect curves (Figure 11). Using the pseudo-inverse, the max across-trial curve variance scales smoothly as $O(\sigma^2)$ (e.g., 3.26×10^{-8} on Synthetic and 1.87×10^{-7} on Housing at $\sigma = 10^{-5}$). A naive rank-blind direct solve instead yields an across-trial variance of 0.309 at the same σ ($9.5 \times 10^6 \times$ higher), producing ill-conditioned curve oscillations. This confirms that the Moore–Penrose pseudo-inverse is necessary and sufficient for stable, interpretable reallocated curves.